LAB 7 – FUZZ TESTING

Course: Software Testing and Automation

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Experiment: Fuzz Testing for Input Validation Bugs

Tools Used:

- Python 3.9.6
- Hypothesis
- Pytest

Objective:

To identify and fix input validation bugs in a Python application using Hypothesis-based fuzz testing.

1) Original buggy application

```
shivanibr@Shivanis-MacBook-Air Lab 7 % python3 student_grades.py
==== Running average_marks ====
Enter marks as CSV (e.g. 78,85,92):
Traceback (most recent call last):
  File "/Users/shivanibr/Downloads/L6_L7_16_student_grades/Lab 7/student_grades.py", line 29, in <module>
    main()
  File "/Users/shivanibr/Downloads/L6_L7_16_student_grades/Lab 7/student_grades.py", line 15, in main
    result = average_marks(user_input)
  File "/Users/shivanibr/Downloads/L6_L7_16_student_grades/Lab 7/student_grades.py", line 2, in average_marks
    marks = [int(m) for m in marks_csv.split(',')]
  File "/Users/shivanibr/Downloads/L6_L7_16_student_grades/Lab 7/student_grades.py", line 2, in <listcomp>
    marks = [int(m) for m in marks_csv.split(',')]
ValueError: invalid literal for int() with base 10: ''
```

2)Fuzz testing failures

```
shivanibr@Shivanis-MacBook-Air Lab 7 % python3 -m pytest test_student_grades.py -v
============================= test session starts ==============================
platform darwin -- Python 3.9.6, pytest-8.4.2, pluggy-1.6.0 -- /Library/Developer/CommandLineTools/usr/bin/python3
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: /Users/shivanibr/Downloads/L6_L7_16_student_grades/Lab 7
plugins: hypothesis-6.141.1, anyio-4.10.0
collected 3 items

test_student_grades.py::test_average_marks_no_crash FAILED               [ 33%]
test_student_grades.py::test_parse_srn_no_crash FAILED                   [ 66%]
test_student_grades.py::test_short_name_no_crash FAILED                  [100%]

================================== FAILURES ===================================
```

3) Bug Fixing and Application Execution

```
shivanibr@Shivanis-MacBook-Air Lab 7 % python3 student_grades.py
==== Running average_marks ====
Enter marks as CSV (e.g. 78,85,92): 78,85,92
Average Marks: 85.0

==== Running parse_srn ====
Enter an SRN (e.g. PES1UG23CS123): PES1UG24CS439
Campus, Year and Dept: ('PES1', 24, 'CS')

==== Running short_name ====
Enter first and last name (e.g. ravi kumar): SHIVANI BR
Short Name: S. Br
```

4)Re-execution of Fuzz Tests

```
shivanibr@Shivanis-MacBook-Air Lab 7 % python3 -m pytest test_student_grades.py -v
============================= test session starts ==============================
platform darwin -- Python 3.9.6, pytest-8.4.2, pluggy-1.6.0 -- /Library/Developer/CommandLineTools/usr/bin/python3
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: /Users/shivanibr/Downloads/L6_L7_16_student_grades/Lab 7
plugins: hypothesis-6.141.1, anyio-4.10.0
collected 6 items

test_student_grades.py::test_average_marks_valid PASSED                  [ 16%]
test_student_grades.py::test_average_marks_invalid PASSED                [ 33%]
test_student_grades.py::test_parse_srn_valid PASSED                      [ 50%]
test_student_grades.py::test_parse_srn_invalid PASSED                    [ 66%]
test_student_grades.py::test_short_name_valid PASSED                     [ 83%]
test_student_grades.py::test_short_name_invalid PASSED                   [100%]

============================== 6 passed in 0.31s ===============================
```